

Codefest 8

Hardware for Artificial Intelligence and Machine Learning (HW4AI)

ECE 410/510

Spring 2026

Overview

This codefest has a single CMAN challenge. No CLLM this week because M3 is due the same Sunday and your time is better spent on synthesis. CMAN must be completed without any AI assistance.

- **Topic:** Address-Event Representation (AER) — bandwidth analysis for event-driven neuromorphic communication.
- **Deliverable** is committed to your GitHub repository by Sunday May 24, 11:59 pm.

CMAN — AER bandwidth analysis

You are designing the off-chip output interface for a spiking neural network (SNN) accelerator. The SNN has $N = 1024$ output neurons, each with a mean firing rate of $f = 50$ Hz (spikes per second). Spikes are communicated to the host using Address-Event Representation (AER): when a neuron fires, the chip emits a packet containing the neuron address plus framing overhead. There is no clock-driven frame; packets are emitted asynchronously as spikes occur. Assume firing is independent across neurons (Poisson-like statistics).

Tasks

1. Compute the mean aggregate spike rate R (spikes/second) for the whole output layer. Show the formula and substitute values: $R = N \times f$.
2. Each AER packet contains a 10-bit neuron address (since $\log_2(1024) = 10$), a 6-bit timestamp, and a 4-bit framing/parity overhead, for 20 bits per packet total. Compute the mean required AER bandwidth B in bits/second and convert to Mbit/s. Show the formula: $B = R \times 20$.
3. Compare your B to each of the standard interfaces from the M1 table: SPI (≤ 50 Mbit/s), I²C (≤ 3.4 Mbit/s), AXI4-Lite (assume 100 Mbit/s effective for a narrow bus). For each interface, state whether it can sustain the mean rate, and identify which is the lowest-complexity interface that suffices.
4. Bursts matter. Suppose stimulus arrives that causes 25% of the 1024 neurons to fire within a 1 ms window (synchronous burst). Compute the peak instantaneous bandwidth required in Mbit/s during that window. Compare to the mean bandwidth from task 2 and identify the burst-to-mean ratio. State whether the interface chosen in task 3 can absorb the burst, or whether buffering is required (and roughly how deep).
5. Frame-based comparison. A conventional (non-AER) readout would sample all 1024 neurons every 1 ms regardless of activity, sending 1 bit per neuron per sample. Compute the frame-based bandwidth in Mbit/s and the AER-to-frame bandwidth ratio at the mean firing rate. State the firing rate $f_{\text{crossover}}$ at which AER and frame-based bandwidths are equal (set them equal and solve for f). Briefly state in one sentence what this implies for when AER is the right choice.

CMAN NO AI

Deliverable:

codefest/cf08/cman_aer_analysis.md (or .pdf/.png) containing

1. mean aggregate spike rate R ;
2. mean AER bandwidth B in Mbit/s with formula and substituted values;
3. interface comparison table (SPI, I²C, AXI4-Lite) with sustain Y/N for each and the chosen lowest-complexity interface;
4. burst-peak bandwidth, burst-to-mean ratio, and buffering decision;
5. frame-based bandwidth, AER/frame ratio at $f=50$ Hz, and crossover firing rate $f_{\text{crossover}}$ with one-sentence implication.